

TemporalGraph-ToM: Evaluating Theory-of-Mind Reasoning with Temporal Belief Graphs

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Extended Abstract

Higher-order theory of mind (ToM), the ability to attribute mental states to others, including their beliefs about others’ beliefs, remains a consistent challenge in large language model reasoning. Existing benchmarks evaluate higher-order belief recursion, such as Hi-ToM [1], interactional information asymmetry in FANToM [2], broad social-cognitive ToM abilities in ToMBench [3], first-person social and counterfactual reasoning in EgoSocialArena [4], and comprehensive cognition-inspired ToM paradigms in CogToM [5].

We introduce TemporalGraph-ToM (TGToM), a benchmark of 1k multi-agent stories that probes higher-order theory of mind with temporal belief graphs. TGToM uses controlled simulations up to seventh-order belief attribution, with information asymmetry from public and private claims, observation asymmetry from interleaved exits, agent reasoning involving deception and cooperation, and higher-order counterfactual scenarios. Evaluation questions span world state, higher-order beliefs, causal beliefs, common knowledge, and counterfactual beliefs.

Probing the limits of higher-order reasoning is relevant to LLM safety, agent safety, human-AI interaction, and reasoning explainability. In evaluations of frontier models, we find a reality-bias (omniscient reasoning) failure mode: models default to the actual world state even when an agent’s belief diverges from it. TGToM evaluates whether models can correctly attribute belief updates under a belief propagation rule and exposes misattribution when models confuse the true world state with an agent’s propagated belief.

Reasoning scaffold. We introduce Temporal Belief Graph (TBG) and compare it with CoT and System 2 prompting. TBG is formalized as a temporal multilayer graph where nodes represent each agent’s state (belief, intent, observation) updated after an event. Beliefs propagate through four directed

link types in two layers: communication links (trusted claim, untrusted claim) and intent links (cooperative, deceptive). Beliefs additionally update directly when an agent observes an event. The node states and link types are generalizable to multiple beliefs, other mental states, event types, and agent interactions.

Temporal Belief Graph

Temporal belief graph is formalized as a temporal multilayer graph where nodes represent each agent’s state (belief, intent, observation) updated after an event. Node state for agent i at time t :

$$s_i^{(t)} = \left(b_i^{(t)}, \iota_i^{(t)}, o_i^{(t)} \right)$$

where:

- $b_i^{(t)}$: belief
- $\iota_i^{(t)}$: intent
- $o_i^{(t)}$: observation

Beliefs propagate through four directed link types in two layers: communication links (trusted claim, untrusted claim) and intent links (cooperative, deceptive). Beliefs additionally update directly when an agent observes an event. Directed link types:

$$e_{j \rightarrow i}^{(t)} \in \{\text{trusted, untrusted, cooperative, deceptive}\}$$

Update rule given event:

$$s_i^{(t+1)} = \begin{cases} f_{\text{obs}}(s_i^{(t)}, E_t) & \text{if observed} \\ f_{\text{trust}}(s_i^{(t)}, E_t) & \text{if trusted} \\ f_{\text{intent}}(s_i^{(t)}, E_t) & \text{if deceptive} \\ s_i^{(t)} & \text{otherwise} \end{cases}$$

Related LLM reasoning scaffolds include temporal belief-state chains, as in TimeToM [6], and graph-structured reasoning, as in Graph of Thoughts (GoT) [7]. Temporal Belief Graph combines the temporal dimension with graph representations for multi-agent belief propagation.

References

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